A multi-base supply vessel planning problem arising in offshore oil and gas operations

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Abstract

Upstream logistics in the Norwegian oil industry includes, among other activities, providing the offshore installations with necessary supplies. These are periodically delivered to the installations by supply vessels from onshore bases. In this paper an integer programming formulation and a large-neighbourhood-search-based heuristic is proposed for a real-world multi-base periodic supply vessel planning problem, which calls for the determination of a repetitive weekly sailing plan for each base, consisting of scheduled voyages, and a common fleet configuration needed to perform these voyages. The advantage of a coordinated planning comes from the economic potential of sharing supply vessels between the bases thus reducing the overall number of vessels needed. Computational results reveal the benefits of a multi-base method as opposed to a single-base approach.

Keywords: maritime transportation; periodic routing; fleet sizing; multiple vehicle use; oil and gas; supply operations; multi-depot problem

1 Introduction

Oil and gas operators use supply vessels to service offshore installations from onshore bases. A prespecified set of offshore installations is served from each onshore base by supply vessels. The assignment of installations to the supply bases is fixed by the government licence regulations. Occasionally an installation may have its different demand types delivered from different supply bases in which case it is served from more than one base. But the assignment of the installation to several bases is still fixed, i.e. the installation may not be served from supply bases other than the ones it is assigned to. Supply vessel expenses constitute a significant cost component in the area of upstream logistics as their daily charter rates may be as high as 80,000 US dollars. Our aim in such a context is to design cost-efficient periodic voyages and schedules (also called sailing plans) for supply vessels as well as respective fleet configuration. This study is based on a real-world

problem faced by Statoil, the largest Norwegian offshore oil and gas operator. The problem where periodic supply vessel voyages and schedules were designed for a single base was considered in Shyshou et al. (2010). The authors referred to the problem as the periodic supply vessel planning problem (PSVPP). In this paper we study an extension of the PSVPP to a case where the planning is simultaneously done for several bases. We refer to the problem as the multi-base periodic supply vessel planning problem (MBPSVPP). All practical constraints valid for the PSVPP also apply in the case of the MBPSVPP, which consists of simultaneously determining a repetitive weekly vessel sailing plan for each base, made up of scheduled voyages, and a common fleet configuration required to perform these voyages. Each scheduled voyage is defined by a start day and a sequence of installations to visit. The objective is to minimize the sum of the vessel charter costs and of the voyage sailing and service costs.

The interest of planning vessel voyages and schedules for several bases in a coordinated fashion comes from the economic potential of sharing supply vessels between the bases. This may lead to an overall smaller number of vessels needed to perform all voyages of the sailing plans, as illustrated in the following example for two supply bases (FFB and FMO). Their uncoordinated weekly sailing plan is depicted in Figure 1. When the planning is done independently for each base four vessels are needed to perform all the voyages and the cost of the sailing plan is NOK 3 427 790. Note that the cost figures we use are based on realistic market values and not on the data provided by Statoil due to confidentiality considerations. The voyages have a bold outline in the figure. Each voyage is preceded by an eight-hour period at the onshore base, when loading and unloading operations take place. In the figure these periods are shaded with diagonal stripes. It can be seen that vessel Symphony starts its third voyage on Saturday at 16:00 (144 hours after the beginning of the week) and finishes it on the Monday of the following week. Also note that vessels TBN and Star each perform one voyage per week. A vessel is said to be *isolated* if it performs only one voyage during the week.

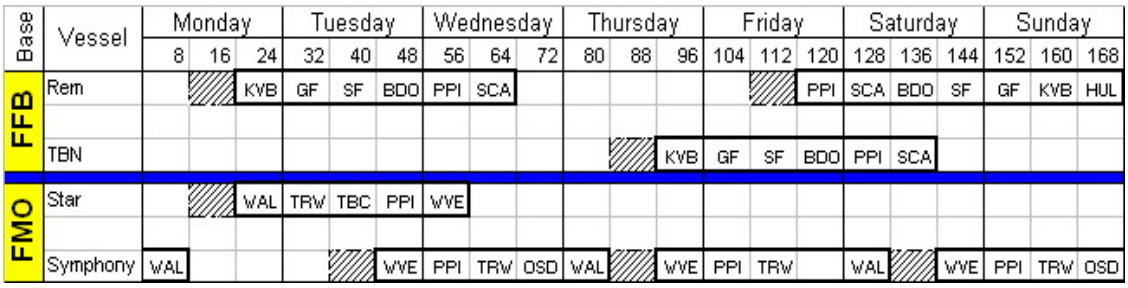

Figure 1: Uncoordinated multi-base weekly sailing plan for bases FFB and FMO

Allowing vessel sharing between two bases leads to a coordinated weekly sailing plan with three vessels displayed in Figure 2. Observe that the reduction in the number of vessels needed to perform all voyages of the two sailing plans was achieved by assigning both voyages previously performed by isolated vessels to only one of them (TBN). The voyage previously performed by Star is reassigned to vessel TBN which now performs two *inter-base* voyages starting at one base and ending at another one. Inter-base voyages have a double-line outline in the figure. Also note that in multi-base schedule all voyages

previously performed by isolated vessels become inter-base voyages. All voyages of vessel Symphony have been reassigned to vessel Star since the charter cost for Symphony is higher than that of Star. The cost of the coordinated schedule is reduced to NOK 2 680 260, yielding a saving of 21,8 %.

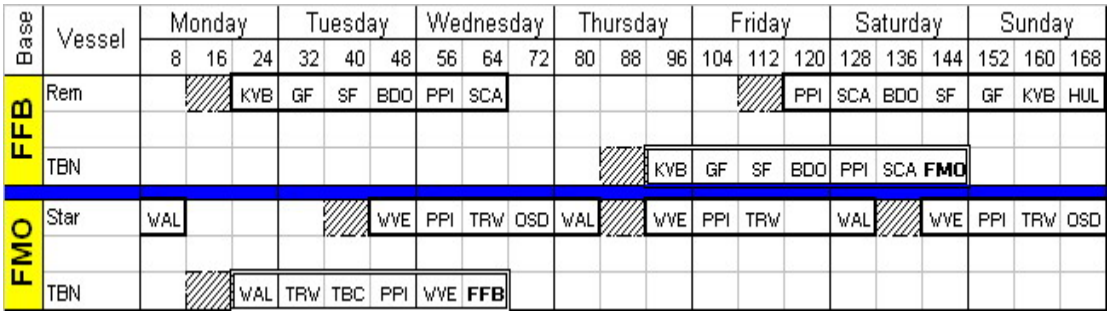

Figure 2: Coordinated multi-base weekly sailing plan for bases FFB and FMO

Inter-base voyages involve indirect movements between bases, i.e. after completing a voyage starting at a given base the vessel performing it has to sail to another base. As the majority of the installations are normally assigned to their nearest base (installations served from several supply bases being an exception), inter-base voyages typically increase sailing costs. It may be possible to reduce the number of inter-base voyages, while keeping the same number of vessels used as in the coordinated schedule, by extending the planning horizon to two weeks and using symmetric schedules. We define two multi-base schedules m_i and m_j to be symmetric if there is an inter-base voyage starting at base k which is first in m_i and last in m_j, and there is an inter-base voyage starting at base $p \neq k$ which is last in m_i and first in m_j. A multi-base sailing plan symmetric to the one of Figure 2 is depicted in Figure 3. This sailing plan is another feasible MBPSVPP schedule, however, the inter-base voyage of TBN starting at FFB is now at the beginning of the week, whereas the inter-base voyage starting at FMO is at the end of the week. The cost of the schedule is also NOK 2 680 260.

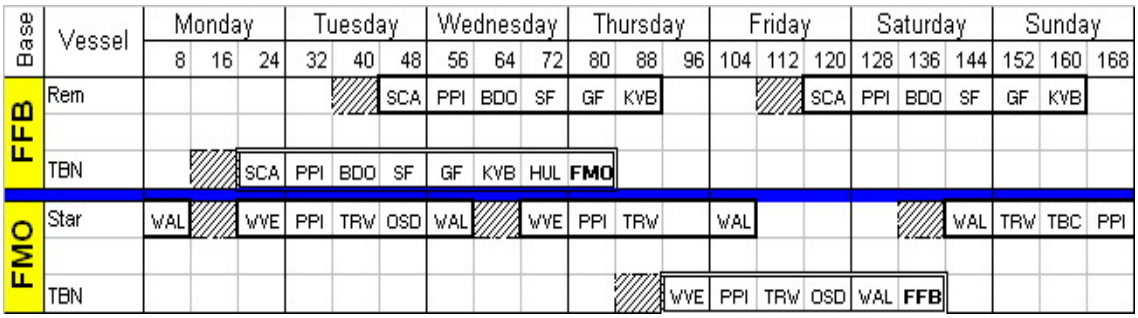

Figure 3: Symmetric multi-base weekly sailing plan

Extending the planning horizon to two weeks by combining two symmetric multi-base weekly sailing plans into a two-week multi-base sailing plan may allow a reduction of one inter-base voyage per week. This can be done by ending the last inter-base voyage of the

first week at base k and starting the first inter-base voyage of the second week from the same base. The same applies to base p for the second and first week, respectively. This is illustrated in the two-week sailing plan shown in Figure 4, which is a combination of the two symmetric schedules of Figures 2 and 3. Vessel TBN starts its first voyage of a two-week time horizon on Monday at base FMO and ends it at base FFB on Wednesday. The second voyage now becomes a regular one starting on Friday and ending on Sunday at FFB as the third voyage also starts at FFB. The third voyage is an inter-base voyage which ends at FMO on Thursday. The fourth voyage now becomes a regular FMO voyage starting on Thursday and ending on Saturday since the first voyage of the next two-week planning horizon starts at FMO. The total number of inter-base voyages during two weeks is now reduced from four to two. The two-week sailing plan cost is NOK 5 359 050, which is slightly less than repeating a coordinated multi-base weekly schedule of Figures 2 and 3 (NOK 5 360 520).

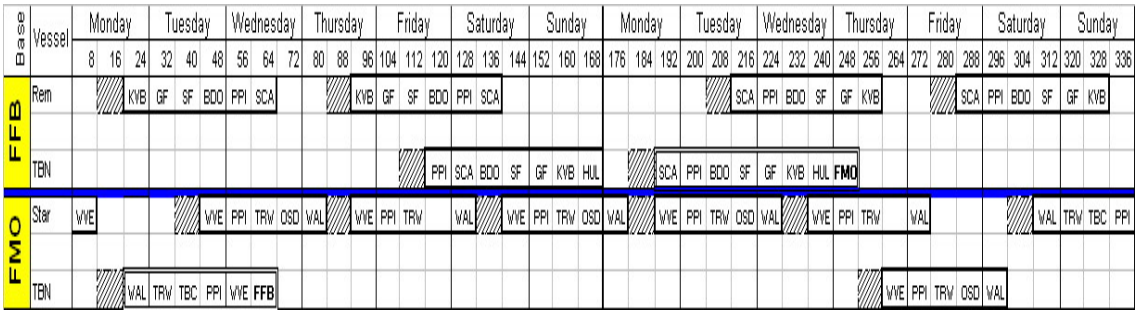

Figure 4: A two-week multi-base sailing plan

The MBPSVPP can be classified as a periodic multi-base vessel routing and fleet sizing problem with time windows and multiple use of vessels. A number of problem aspects stand out from the classification, namely multi-depot periodic routing with time windows and inter-depot routes, fleet sizing and mix, and multiple use of vehicles during the multi-period planning horizon. Most of these aspects are surveyed in Shyshou et al. (2010). Here we focus on the periodic multi-base aspect with inter-base voyages.

The sources of the multi-depot periodic vehicle routing problem (MDPVRP) are quite scarce. Angelelli and Speranza (2002) have developed a heuristic for a periodic vehicle routing problem in which replenishments at intermediate facilities are allowed. Their solution method is based on the tabu search heuristic of Cordeau et al. (1997). An exact method for the MDPVRP was proposed by Mingozzi (2005). It uses variable pricing to reduce the set of variables to solvable proportions. A recent study of the multi-depot vehicle routing problem with inter-depot routes (MDVRPI) was performed by Crevier et al. (2007). The authors formulate the MDVRPI as an integer linear program and develop a heuristic algorithmic approach, based on a decomposition of the problem into MDVRP, VRPs and inter-depot subproblems. A tabu search heuristic is then applied to each subproblem and the resulting solutions are combined to obtain a solution to MDVRPI. Direct comparisons of solution method for the MBPSVPP to methods developed for other problems in the literature are difficult to make due to presence of periodicity and multiple use of vessels aspects in the MBPSVPP.

The purpose of this paper is to propose a large-neighbourhood-search-based heuristic for the MBPSVPP. The remainder of the paper is organized as follows. A mathematical formulation for the MBPSVPP is presented in the next section, followed by the description of the heuristic. We then proceed to computational experiments on real-world instances. Finally, conclusions are drawn.

2 Mathematical model

In this section we present the extension of the PSVPP formulation proposed in Shyshou et al. (2010) for the MBPSVPP. Let G be the set of supply bases defining the MBPSVPP instance, let N be the set containing all offshore installations assigned to all bases and let V be the set of all available supply vessels. Further, let P be the set representing the weeks of the planning horizon (normally one or two weeks), let T_p be a circularly ordered set representing the days of week p ($p \in P$), and let $\overline{T}$ be a circularly ordered set representing the days of $|P|$ consecutive weeks of the planning horizon ($\overline{T} = \cup_{p \in P} T_p$). Also let R_j be the set of pregenerated cheapest feasible voyages (including inter-base voyages) vessel $j \in V$ may sail. Let L be a set of all possible voyage durations measured in days and let F be a set of all possible installation visit frequency values. Additionally, let $R_{jl} \subseteq R_j$ be a set of pregenerated cheapest feasible voyages for vessel $j \in V$ with a duration of l days ($l \in L$) and let N_r be a subset of the installations requiring r visits per week ($r \in F$).

The formulation also uses the following notation: c_j is the charter cost for using vessel $j \in V$ over the entire planning horizon of length $\left|\overline{T}\right|$; b_{tg} is an upper bound on the number of vessels leaving the onshore base $g \in G$ on day $t \in \overline{T}$; c_{jk} is the sailing and service cost associated with vessel $j \in V$ sailing voyage $k \in R_j$; s_i is the required number of visits per week for installation $i \in N$; a_{ijkg} are binary constants ($i \in N, j \in V, k \in R_j, g \in G$) equal to 1 if and only if vessel j visits installation i on voyage k starting from base g. Since a prespecified set of installations is assigned to each base g, a_{ijkg} will be zero for all those $i \in N$ which are not assigned to g. The parameter d_{jk}, rounded up to the nearest integer, represents the duration (in days) of voyage k sailed by vessel j ($j \in V, k \in R_j$); f_{jp} is the number of days vessel j is available for during week p ($j \in V, p \in P$). Define $0 \leq h_r \leq \left|\overline{T}\right| / |P|$ to be the length of an auxiliary sub-horizon for the installations whose visit frequency is r. Also define $\overline{p}_r$ and $\underline{p}_r$ as an upper and a lower bound on the number of visits during the sub-horizon of length h_r an installation whose visit frequency is r ($r \in F$) should receive. We also introduce binary constants e_{jkg} equal to one if and only if voyage k sailed by vessel j ends at base g ($k \in R_j, j \in V, g \in G$) and q_{jg} equal to one if and only if vessel j is located at base g at the beginning of the planning horizon ($j \in V, g \in G$).

The decision variables of the model are:

$$x_{jktg} = \begin{cases} 1, \text{ if vessel } j \text{ starts voyage } k \text{ on day } t \text{ from base } g \ (j \in V, k \in R_j, t \in \overline{T}, g \in G) \\ 0, \text{ otherwise} \end{cases}$$

$$y_j = \begin{cases} 1, \text{ if vessel } j \text{ is used } (j \in V) \\ 0, \text{ otherwise.} \end{cases}$$

$$z_{jtg} = \begin{cases} 1, \text{ if vessel } j \text{ is either located at base } g \text{ on day } t \\ \text{or is performing a voyage during day } t \text{ which ends at base } g \ (j \in V, t \in \overline{T}, g \in G) \\ 0, \text{ otherwise.} \end{cases}$$

The mathematical model is then:

$$\text{minimize} \sum_{j \in V} c_j y_j + \sum_{j \in V} \sum_{k \in R_j} \sum_{t \in \overline{T}} \sum_{g \in G} c_{jk} x_{jktg} \tag{1}$$

subject to

$$\sum_{j \in V} \sum_{k \in R_j} \sum_{t \in T_p} \sum_{g \in G} a_{ijkg} x_{jktg} \geq s_i, i \in N, p \in P \tag{2}$$

$$\sum_{k \in R_j} \sum_{t \in T_p} \sum_{g \in G} d_{jk} x_{jktg} - f_{jp} y_j \leq 0, j \in V, p \in P \tag{3}$$

$$\sum_{j \in V} \sum_{k \in R_j} x_{jktg} \leq b_{tg}, t \in \overline{T}, g \in G \tag{4}$$

$$\sum_{k \in R_j} x_{jktg} \leq z_{jtg}, j \in V, t \in \overline{T}, g \in G \tag{5}$$

$$\sum_{k \in R_{jl}} \sum_{g \in G} x_{jktg} + \sum_{k \in R_j} \sum_{g \in G} \sum_{q=1}^{l-1} x_{jk,(t+q) \bmod |\overline{T}|,g} \leq y_j, j \in V, t \in \overline{T}, l \in L \tag{6}$$

$$\underline{p}_r \leq \sum_{j \in V} \sum_{k \in R_j} \sum_{h=0}^{h_r} \sum_{g \in G} a_{ijkg} x_{jk,(t+h) \bmod |T_p|,g} \leq \overline{p}_r, i \in N_r, t \in T_p, r \in F, p \in P \tag{7}$$

$$z_{j1g} = q_{jg} y_j, j \in V, g \in G \tag{8}$$

$$z_{j,(t+1) \bmod |\overline{T}|,g} \leq z_{jtg} + \sum_{k \in R_j} (e_{jkg} - 1) x_{jktg} + \sum_{b \in G \setminus \{g\}} \sum_{k \in R_j} e_{jkg} x_{jktb}, j \in V, t \in \overline{T}, g \in G \tag{9}$$

$$y_j \in \{0,1\}, j \in V \tag{10}$$

$$x_{jktg} \in \{0,1\}, j \in V, k \in R_j, t \in \overline{T}, g \in G \tag{11}$$

$$z_{jtg} \in \{0,1\}, j \in V, t \in \overline{T}, g \in G. \tag{12}$$

The objective function (1) minimizes the sum of vessel charter costs and of voyage sailing and service costs. Constraints (2) ensure that each installation receives the required number of visits per week. Constraints (3) limit the use of a vessel to the number of days for which it is available in each week, whereas constraints (4) guarantee that an upper bound on the number of vessels leaving the onshore base on a given day is not exceeded. Constraints (5) state that a vessel can only start a voyage from a given base if it is located at this base. Constraints (6) mean that if a used vessel starts a voyage of l-day duration on day t from any base, it cannot start another voyage from any base on days from t to $t + l - 1$. Constraints (7) ensure the uniformity of departures to the installations, depending on their visit frequency, during each week of the planning horizon. For example let us consider $r = 3$: the corresponding parameter values are $h_3 = 3$, $\overline{p}_3 = 2$ and $\underline{p}_3 = 1$, which results in a constraint that requires at least one departure and no more than two departures every three days for an installation whose visit frequency is three. Since we aim to identify a repetitive sailing plan for $|P|$ weeks, constraints (8) specify that vessel locations at the beginning of each planning horizon should be the same for the vessels which are used. Constraints (9) guarantee the continuity of vessel locations. These constraints state that

a vessel becomes unavailable at one base and becomes ”located” at another base when it starts performing an inter-base voyage. Otherwise, the location of the vessel does not change. Constraints (10) – (12) impose the binary requirements on the variables.

It should be noted that the value of this formulation is mostly descriptive as it cannot be readily applied to realistic size instances. For example, in the single-base problem (see Shyshou et al., 2010), the largest instance size that could be solved exactly by enumerating all feasible voyages (Halvorsen-Weare et al., 2009) contained 12 installations, five vessels and 51 visit. The large neighbourhood search heuristic developed for this problem yielded optimal or quasi-optimal solutions on all instances of the test-bed. The size of the present problem is much larger than the single-base case because voyages can now start and end at any one of several bases. We therefore advocate the use of a heuristic which we now describe.

3 A large-neighbourhood-search-based heuristic for the MBPSVPP

In this section we first outline the heuristic approach conceptually. We then proceed to a step-by-step description of the algorithm.

3.1 Conceptual description of the heuristic

Our solution method is an extension of the large neighbourhood search (LNS) heuristic developed by Shyshou et al. (2010). An MBPSVPP instance consists of $n = |G|$ single-base PSVPP instances $I_1, I_2, ..., I_n$. The initial stage of the heuristic solution approach consists of applying the LNS heuristic of Shyshou et al. (2010) to each PSVPP instance defining the MBPSVPP instance. However, now in addition to finding the cheapest sailing plan, we are also interested in a set of weekly sailing plans heterogeneous with respect to starting day of the voyages performed by isolated vessels. We have developed an extra procedure which, when applied to a feasible solution, attempts to isolate a vessel through one-way or two-way (swap) reassignments of voyages to vessels. Denote by $W_1, W_2, ..., W_n$ the sets of heterogeneous weekly sailing plans for the respective PSVPP instances $I_1, I_2, ..., I_n$. The maximum cardinality of $W_k, k = 1, ..., n$ is six since we have vessel departures from bases every day, except Sunday. Having produced these sets, we proceed to combining individual weekly sailing plans into a multi-base sailing plan consisting of n single-base sailing plans with a vessel shared between them. Each single-base schedule defining the multi-base sailing plan is post-optimized using LNS heuristic for the single-base PSVPP of Shyshou et al. (2010). We store in a set of diverse multi-base schedules M every combined multi-base schedule yielding a reduction in the number of vessels used. We then perform pairwise combinations of symmetric multi-base sailing plans in which the same vessel is shared between the bases (if such pairs exist) into a two-week n-base sailing plan. If a pair of symmetric multi-base schedules is identified, the last inter-base voyage of each multi-base schedule is made regular (starting and ending at the same base), and each single-base weekly sailing plan, whose number of inter-base voyages has been reduced, is then post-optimized by means of the LNS heuristic for the single-base PSVPP.

The multi-base heuristic returns the best schedule of each type: 1) uncoordinated weekly multi-base sailing plan; 2) a multi-base sailing plan with a shared vessel; 3) a two-week multi-base sailing plan with a reduced number of inter-base voyages.

3.2 Detailed description of the heuristic framework

We now proceed to a detailed description of the heuristic solution framework for the MBPSVPP. We start with the description of an auxiliary procedure to isolate a vessel designed specifically for the MBPSVPP. We then provide a step-by-step description of the algorithm.

3.2.1 Isolating a vessel

We have developed a special procedure for isolating a vessel in a weekly sailing plan. Recall that a solution to the PSVPP consists of a collection of voyages and of an assignment of vessels to these voyages. Also recall that a vessel is isolated if it performs only one voyage during the week. The procedure consists of two consecutive loops: 1) swap reassignments and 2) one-way reassignments. A swap reassignment assigns voyage r_1 of vessel v_1 to vessel v_2, and voyage r_2 of vessel v_2 to v_1. A swap reassignment is called *improving* if it increases the number of consecutive days a vessel is used and does not decrease it for the other vessel. The loop terminates when no improving swap reassignment is possible. The intuition behind the loop of swap reassignments is to concentrate the voyages performed by a given vessel close to each other so that there is only one period of idle time. This single idle period offers a better voyage reassignment potential than several idle periods scattered around the week. In the second loop we consider each vessel in turn and try to isolate it by reassigning all but one of its voyages to different vessels. The loop terminates with a reassignment leaving an isolated vessel (if any has been identified) of largest achievable capacity. We apply this procedure within the LNS heuristic whenever a feasible single-base solution is identified after all local improvement procedures have been performed.

3.2.2 Step-by-step description of the heuristic

Before proceeding to the description of the heuristic we introduce some notation. Let $I = \{ I_1, I_2, ..., I_n\}$ be the set of PSVPP instances defining the MBPSVPP instance. Denote by w_k be an arbitrary feasible single-base weekly sailing plan for instance $I_k, k = 1, ..., n$ and let $c(w_k)$ be its cost. Let w_k^* be the incumbent single-base weekly sailing plan of minimum cost for instance I_k. Further, denote by w_k^d be a feasible weekly sailing plan for I_k with an isolated vessel whose voyage starts on day d. Recall that W_k denotes a set of weekly sailing plans for instance I_k heterogeneous with respect to starting day of the voyage performed by isolated vessel. To control the quality of sailing plans in W_k we introduce a parameter δ. The steps of the heuristic are then as follows.

Step 1 *Generation of single-base schedules*

For each PSVPP instance I_k, $k = 1, ..., n$ of set I

- Set $W_k = \emptyset$.
- Apply the LNS heuristic for ρ restarts and for η iterations in each restart identifying and updating w_k^*. Define lexicographic ordering u: 1) number of vessels used (the smaller the better); 2) capacity of the isolated vessel (the larger the better); 3) cost of the sailing plan (the smaller the better). During the LNS update W_k in the following way.
  - **If** during the search w_k^d is found, such that $(c(w_k^d) - c(w_k^*))/c(w_k^*) < \delta$, and there is no sailing plan with an isolated vessel whose voyage starts on day d in W_k, then set $W_k = W_k \cup \{w_k^d\}$.
  - **Else if** there is a previously found $\widetilde{w_k^d}$ in W_k and currently found w_k^d is better than $\widetilde{w_k^d}$ in terms of u, set $W_k = (W_k \setminus \{\widetilde{w_k^d}\}) \cup \{w_k^d\}$.
  - **End if**
- Do a post-processing of W_k removing all sailing plans using more vessels than in w_k^*.

End for

Step 2 *Generation of coordinated weekly multi-base schedules with a shared vessel*

Proceed to evaluating all possible combinations of n single-base weekly sailing plans from W_k, $k = 1, ..., n$ into one n-base weekly sailing plan. Total number of combinations is $\prod_{k=1}^{n} |W_k|$. Let us denote an arbitrary multi-base sailing plan by m and a set of diverse coordinated multi-base schedules with a shared vessel by M. Set $M = \emptyset$.

- **For** each uncoordinated multi-base weekly sailing plan m_i, $i = 1, ..., \prod_{k=1}^{n} |W_k|$:
  - Let D_i be a set of voyages performed by isolated vessels in single-base schedules defining m_i. Sort D_i in increasing order of the voyage start times. As a result the first voyage of D_i would be the earliest, whereas the last voyage of D_i would be the latest. Also let Q be the set of vessels performing voyages of D_i.
  - **For** each voyage $j \in D_i$:
    - End voyage j at the base where voyage $(j+1) \bmod |D_i|$ starts thus making it an inter-base voyage.

    - Apply the intra-voyage optimization procedure of Shyshou et al. (2010) to voyage j.
  - **End for**
  - **If** all voyages in D_i are non-overlapping, i.e. can be performed by one vessel:
    - Assign a vessel $q^* \in Q$ of smallest feasible capacity to perform all voyages of D_i.
    - Set $Q = Q \setminus \{q^*\}$.
    - Liberate all remaining vessels in Q.
    - Apply LNS (without local improvement procedures other than reassignment of vessels to voyages) to each single-base weekly sailing plan defining m_i.
    - Set $M = M \cup \{m_i\}$.
  - **End if**

- **End for**

Step 3 *Generation of two-week multi-base schedules with reduced number of inter-base voyages*

Attempt to combine all possible couples of set M (there are $|M| \times |M-1|/2$ in total) into two-week n-base sailing plans with reduced number of inter-base voyages.

- **For** $i = 1, ..., |M| - 1$
  - **For** $j = 2, ..., |M|$
    - **If** $j > i$ and m_i is symmetric to m_j and the shared vessel is the same for both schedules
      - End the last inter-base voyage of D_i in week t at the base from which the voyage was started (say base k), as the first inter-base voyage of D_j in week $t+1$ starts from base k (since m_i and m_j are symmetric). Apply the intra-voyage optimization procedure on the last voyage of D_i, which now becomes a regular voyage thereby reducing the number of inter-base voyages by one.
      - End the last inter-base voyage of D_j in week $t+1$ at the base from which the voyage was started (say base p), as the first inter-base voyage of D_i in week $t+2$ starts from base p (again due to symmetry between m_i and m_j). Apply the intra-voyage optimization procedure on the last voyage of D_j, which now becomes a regular voyage thereby reducing the number of inter-base voyages by one.
      - Apply LNS (without local improvement procedures, only applying reassignment of vessels to voyages) to the single-base weekly sailing plans in which the number of inter-base voyages have been reduced.

- If the current two-week multi-base schedule is of smaller cost than the best incumbent two-week sailing plan, store the current schedule as the best one.

- **End if**

- **End for**

- **End for**

Return the best schedule of each type (if any has been identified):

1) uncoordinated multi-base weekly sailing plan, which is simply a collection of best single-base schedules without sharing a vessel;

2) coordinated multi-base weekly sailing plan (a multi-base schedule of smallest cost belonging to M);

3) two-week multi-base sailing plan with a reduced number of inter-base voyages.

4 Computational experiments

We evaluate our heuristic on instances derived from real-world data. These involve three bases (CCB, FFB and FMO) with a number of installations varying from 3 to 14 for each base. The value of the parameter δ, controlling the quality of solutions in sets of single-base weekly schedules heterogeneous with respect to the starting day of the voyage performed by the isolated vessel, was chosen to be 0.15. The number of restarts ρ was set to 20. The number of LNS iterations η at each restart depends on the number of installations defining the PSVPP instance: 1) up to four installations: five iterations; 2) five to eight installations: 10 iterations; 3) nine to 11 installations: 75 iterations; 4) 12 installations or more: 200 iterations. The number of post-optimization LNS iterations is also instance-size dependent (for both multi-base stage and two-week multi-base stages of the heuristic): 1) up to four installations: five iterations; 2) five to eight installations: 10 iterations; 3) nine to 11 installations: 50 iterations; 4) 12 installations or more: 75 iterations.

We restrict ourselves to pairwise combinations of the weekly sailing plans since combining more than two bases is impractical in our application due to restrictions on voyage durations. Recall that all voyages should be two to three days long. As a result we compare the costs of best two-base coordinated (vessel sharing allowed) weekly sailing plans with the costs of best uncoordinated (no vessel sharing) two-base weekly sailing plans in Table 1. The selection of instance couples for which we report results is based on their synergy potential. To identify these couples, preliminary tests were performed and single-base instances where vessels could be successfully isolated were identified. Resulting couples vary in size and contain between 10 and 23 installations. The first column of the table contains the instance names. An instance name consists of a pair of base names with the respective number of installations for each base, e.g. "CCB-4_FMO-6" means that the instance involves base CCB with four installations and base FMO with six installations. The second and the third columns reveal the costs of uncoordinated multi-base weekly sailing plans decomposed into charter and sailing components, whereas total

costs are given in the fourth column. Recall that although we are not using actual cost data of Statoil due to confidentiality considerations, our charter and sailing and service cost estimates are based on realistic market values. Percentage savings in charter, sailing and total costs are reported in the three last columns, respectively.

Instance	**Uncoordinated plan cost components**			**Coordinated plan savings (%)**		
	Charter	**Sailing**	**Total**	**Charter**	**Sailing**	**Total**
CCB-4_FMO-6	2 964 000	565 470	3 529 470	25.47	0.48	21.47
CCB-4_FFB-7	2 915 000	540 400	3 455 400	25.90	0.60	21.95
FMO-6_FFB-7	2 905 000	522 790	3 427 790	25.78	–0.28	21.81
CCB-4_FFB-11	3 657 000	703 880	4 360 880	20.65	0.63	17.41
CCB-4_FMO-12	4 574 000	978 980	5 552 980	18.36	–0.12	15.11
FMO-6_FFB-11	3 647 000	685 780	4 332 780	20.54	–0.16	17.26
FFB-7_FMO-12	4 515 000	933 960	5 448 960	18.60	–4.10	14.71
FFB-11_FMO-12	5 257 000	1 103 860	6 360 860	15.98	–1.03	13.03
Average				**21.41**	**–0.50**	**17.84**

Table 1: Comparison between coordinated and uncoordinated multi-base sailing plans

Observe that using one vessel less (since we are considering two bases) to perform the multi-base weekly sailing plan leads to a considerable reduction in total cost. The savings vary between slightly more than 13% and almost 22%. Also note that allowing vessel sharing and inter-base voyages leads to reduction in sailing costs for instances "CCB-4_FMO-6", "CCB-4_FFB-7" and "CCB-4_FFB-11". This happens when the bases are located close to each other and the installations are not assigned to its nearest base. However, for the remaining five instances the savings in sailing costs are negative.

In Table 2 we analyze the option of introducing a multi-base two-week sailing plan with a reduced number of inter-base voyages. This may help to reduce sailing costs for instances where coordinated multi-base planning leads to their increase. The second column contains the costs of such plans, the third and fourth columns report savings decomposed into charter and sailing components, whereas the fifth column shows total cost savings with respect to repeated best coordinated weekly plan. Total computation times for all stages of the heuristic are given in the last column.

It can be observed that extending the planning horizon to two weeks and using sailing plans with a reduced number of inter-base voyages is beneficial for two instances out of eight ("FMO-6_FFB-7" and "FMO-6_FFB-11"). For instances in which the introduction of inter-base voyages led to a decreased sailing cost ("CCB-4_FMO-6", "CCB-4_FFB-7" and "CCB-4_FFB-11") the reduction of number of inter-base voyages certainly leads to a cost increase. For larger instances ("FFB-7_FMO-12" and "FFB-11_FMO-12") relatively large incremental total cost can be explained by the inability of the heuristic to find symmetric multi-base weekly schedules of solution quality similar to the best coordinated multi-base weekly schedule. Additionally, in a two-base two-week schedule for instance "FFB-7_FMO-12" the shared vessel is also of larger capacity (and, hence, more expensive) than the one in a coordinated two-base weekly sailing plan. Clearly, for instances where extending the planning horizon to two weeks does not offer cost savings, applying

Instance	Two-week plan with reduced number of voyages	Savings with respect to repeated coordinated weekly plan (%)			Total CPU Time (s)
	Total cost	Charter	Sailing	Total	
CCB-4_FMO-6	5 546 230	0	–0.20	–0.05	163.8
CCB-4_FFB-7	5 399 470	0	–0.50	–0.10	327.1
FMO-6_FFB-7	5 359 050	0	0.10	0.03	369.1
CCB-4_FFB-11	7 207 160	0	–0.30	–0.06	12 855.6
CCB-4_FMO-12	9 435 260	0	–0.36	–0.07	7 409.9
FMO-6_FFB-11	7 168 900	0	0.06	0.01	8 243.4
FFB-7_FMO-12	9 490 450	–3.43	2.88	–2.11	7 327.8
FFB-11_FMO-12	11 132 270	0	–3.04	–0.61	14 296.0
Average		**–0.43**	**–0.17**	**–0.37**	**6 374.1**

Table 2: Introducing two-week multi-base sailing plans with a reduced number of inter-base voyages

a coordinated weekly plan is a preferable option.

5 Conclusion

We have considered a multi-base periodic supply vessel planning problem. The problem calls for the simultaneous determination of scheduled voyages from each base and the common fleet configuration to perform these voyages. Supply vessels are a costly resource and an important advantage of a coordinated multi-base planning is the ability to share vessels between the bases, which leads to a reduction in number of vessels needed to perform all voyages. We proposed a mathematical model for the problem and developed a large-neighbourhood-search heuristic. The heuristic can be used as a decision support tool for single-base and multi-base supply vessel planning. Our computational results, based on instances derived from real-world data, show that a coordinated multi-base approach is helpful in identifying vessel sharing potential and reveal cost savings of around 18% on average for problem instances where vessels can be successfully isolated.

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